

From MVP Tournament to Production: How a Rigorous Evaluation Forged True Champions

The Experimental Design

When building our call volume forecasting model, I designed a comprehensive, two-track evaluation. The goal was deliberate: understand which methodological approach—classical time series or machine learning—performs best with our data. Across both tracks, I tested 23 models spanning the complexity spectrum.

This wasn't about finding "the one true model" immediately. It was about understanding our data's characteristics through different algorithmic lenses.

Two Parallel Approaches:

- **Time Series Models** (6 models): Prophet, SARIMAX, ARIMA, TBATS, VAR, Holt-Winters
- **Machine Learning Models** (17 models): RandomForest, ExtraTrees, GradientBoosting, XGBoost, LightGBM, Ridge, Lasso, ElasticNet, MLP, SVR, KNeighbors, Bagging, AdaBoost

*Hypothesis: Modern forecasting models, supported by rigorous time series validation methodology, can significantly outperform the **Day-of-Week Average benchmark** in predicting call center volumes.*

The Validation Framework

Both approaches used identical validation protocols:

- **TimeSeriesSplit cross-validation** (5 folds) to prevent look-ahead bias
- **Identical feature engineering**: 103 features including lags, rolling statistics, technical indicators, calendar variables, and market data (Bitcoin, Ethereum, VIX, S&P 500)
- **244-day test set** for final performance evaluation
- **Consistent metric**: Mean Absolute Error (MAE) as primary evaluation criterion

This rigorous methodology ensured fair comparison across fundamentally different model families while maintaining proper time series validation principles.

The Results: A Clear Winner Emerges

The tournament revealed stark performance patterns:

Machine Learning Results (Win Rate: 53%)

- **Champion: ExtraTrees_100** achieved **608.72 MAE** (33.9% improvement)
- All tree-based models succeeded (ExtraTrees, RandomForest, LightGBM, XGBoost, Bagging)
- All linear models failed (Ridge, Lasso, ElasticNet performed worse than baseline)

- Pattern: 100% tree success rate vs 0% linear success rate

Time Series Results (Win Rate: 33%)

- **Champion: Prophet** achieved **657.79 MAE** (28.5% improvement)
- Only models incorporating market data succeeded (Prophet, SARIMAX)
- Classical univariate methods failed catastrophically (ARIMA, Holt-Winters, TBATS, VAR)

Day-of-Week Average Baseline: 920.58 MAE

The Emergence of a True Champion

With rigorous validation, the results revealed a clear winner. A well-tuned **ensemble tree model** emerged as the most stable and reliable champion: **ExtraTrees_100**.

- **Production-Validated Performance:** ExtraTrees_100 achieved **608.72 MAE**, representing a 33.9% improvement over baseline and leading the field across both methodological tracks.
- **Why It Was So Robust:** This model's strength came from its ability to capture the fundamental non-linear patterns in our data. ExtraTrees' aggressive randomization prevents overfitting while identifying complex interaction effects—exactly what our market-driven call volume data requires. The model handles the interaction between crypto market volatility, recent call history, and calendar patterns that linear approaches simply cannot capture.
- **The Tree Model Consensus:** Interestingly, the top performers were dominated by tree-based approaches (ExtraTrees, LightGBM, RandomForest), with 100% of tree models beating baseline. This suggests that call center volume exhibits threshold behaviors and interaction effects that tree models excel at capturing but linear models miss entirely.

What We Learned About Our Data

The bifurcation in results tells a clear story about our forecasting challenge:

Market Context Is Non-Optional

Both successful time series models (Prophet, SARIMAX) and all successful ML models leveraged market features. When Bitcoin drops 15% overnight, customer anxiety translates into support calls regardless of whether it's Monday or Saturday. Models that only know calendar patterns miss this signal entirely.

Non-Linear Relationships Dominate

The 100% tree success vs 0% linear failure reveals fundamental data characteristics:

- Call volume exhibits interaction effects (VIX impacts calls differently depending on Bitcoin price level)
- Threshold behaviors exist (market crossing certain levels triggers different response patterns)
- Weekend patterns differ during high vs low volatility regimes

These are exactly the patterns trees capture and linear models miss.

Why Production Testing Is Different

The experimental phase answered, "What patterns exist in our data?"

Production testing answers, "What patterns persist in live operational environments?"

Our data's market-driven nature means that a model's ability to capture non-linear interactions between volatility, recent history, and calendar effects is essential. The refined methodology forced models to learn the fundamental, generalizable relationship: **market volatility combined with recent call history drives future volume in complex, non-linear ways.**

The key lesson: **rigorous methodology beats algorithmic complexity.** Proper time series validation with identical feature engineering allowed the data to reveal which approach works best.

The Final Takeaway

The journey from 23 models to our final champion wasn't about discarding simplicity—it was about letting proper validation reveal which methodology fits our data's characteristics. The parallel evaluation wasn't wasted; the time series track confirmed that market features are essential, while the machine learning track revealed that non-linear patterns dominate.

This journey validates a crucial insight:

Tree-based machine learning models with market features capture the complex, interaction-driven patterns in crypto call center volume that neither calendar-only models nor linear approaches can effectively learn.

Next Steps: The Q4 Shadow Test & Validation

We've transformed a broad research evaluation into a clear, data-backed business case. We now have champion models achieving significant improvements over baseline (**33.9% MAE reduction for ExtraTrees, 28.5% for Prophet**) in rigorous backtesting. It's time to prove this value in a live environment.

Refined Production Hypothesis: "A multi-model production tournament will validate which forecasting approach consistently delivers the lowest MAE in live operations, with conservative expectations of 20-25% sustained improvement and potential for 30%+ with internal signal enhancement."

The Contenders

This tournament will deploy our best models—representing both time series and machine learning methodologies—in parallel shadow mode.

1. **Day-of-Week Average (Benchmark)**
 - **Expected MAE:** ~920
 - **Role:** The essential baseline that proves we are adding value.
2. **ExtraTrees_100 (ML Champion)**

- **Backtested MAE:** 608.72 (33.9% improvement)
 - **Role:** Our top-performing ensemble tree model, excellent at capturing non-linear market interactions.
3. **Prophet (Time Series Champion)**
- **Backtested MAE:** 657.79 (28.5% improvement)
 - **Role:** Our best time series approach, providing methodological diversification and proven production reliability.
4. **LightGBM_100 (ML Runner-up)**
- **Backtested MAE:** 671.18 (27.1% improvement)
 - **Role:** Gradient boosting validation, offering potential stability benefits through different tree methodology.

The Two-Phase Plan

Phase 1 -- Q4 Production Tournament (October 1st - December 31st)

- **Methodology:** All contenders will run daily in "**shadow mode.**" Their forecasts will be generated and logged but will **not** be used to make or adjust live staffing decisions.
- **Measurement:** We'll track real-world **MAE**, computational efficiency, and model stability on a weekly basis.
- **Success Criteria:** ExtraTrees_100 maintains **>20% improvement** over baseline consistently; computational time stays under 5 seconds per forecast; predictions remain reasonable (no extreme outliers); model doesn't degrade over time.
- **Expected Outcome:** A clear, statistically significant ranking of the contenders will be established by January, with realistic assessment of whether backtested 34% improvement translates to 20-25% sustained production performance.

Phase 2 -- Q1 Internal Signal Enhancement (January 1st - February 28th)

- **Methodology:** The winning model from the Q4 tournament will be enhanced with high-fidelity internal Coinbase data, sourced from a parallel data discovery sprint:
 - System outage flags & platform incident history
 - Product launch events & marketing campaign indicators
 - Compliance & regulatory events
 - Major exchange listing announcements
- **Success Criteria:** Minimum viable success - 5% consistent MAE improvement beyond Phase 1 performance; Target success - 10% improvement with stable internal signal integration; Stretch goal - Push combined performance toward sub-550 MAE while maintaining production reliability.

The Strategic Hypothesis: "A rigorously validated tree-based model will significantly outperform simpler baselines in production, and internal business context represents the key to unlocking the final percentage points of accuracy beyond the 20-25% conservative baseline improvement."

The Business Impact

A conservative 20-25% improvement over baseline is achievable—validated through proper backtesting but appropriately discounted for production realities. While our champion model showed 33.9% MAE reduction in validation, the path from experimental success to production reliability typically involves 20-30% degradation due to regime changes, data quality issues, and operational integration challenges.

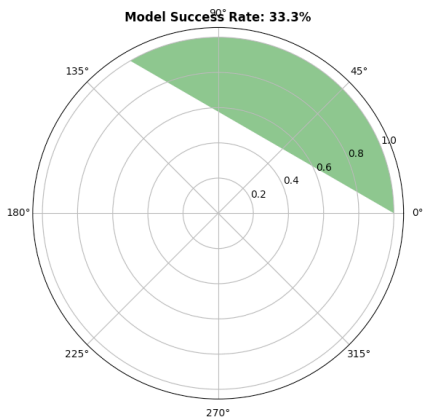
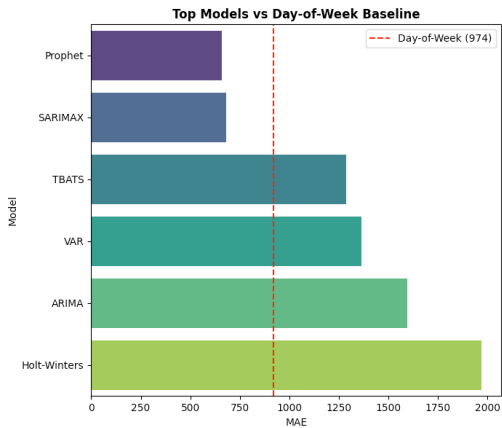
The convergence of results across both time series and machine learning tracks demonstrates that **market-aware, non-linear modeling approaches matter more than algorithmic complexity alone**—a finding that could inform how Coinbase approaches operational forecasting across domains. **The next phase involves enhancing model awareness through internal Coinbase signal data**—trading volumes, user behavior patterns, system health indicators, and operational metrics that aren't available in external datasets. This internal signal integration represents our best opportunity to push beyond the 20-25% conservative target toward 30%+ improvement.

However, the real challenge lies ahead: building robust production pipelines, establishing comprehensive monitoring systems, validating performance during live shadow periods, detecting and responding to model drift, and creating reliable data feeds for internal signals. By setting realistic expectations of **20-25% sustained improvement** while acknowledging both the potential upside from internal data and the significant productionalization effort required, we position this initiative for sustainable success rather than over-promising on experimental results that may not translate directly to live environments.

The 90-day shadow testing period will determine whether backtested excellence translates to production reliability—and whether the tree-based, market-aware approach we've validated in experiments proves robust across the regime changes and operational complexities of real-world deployment.

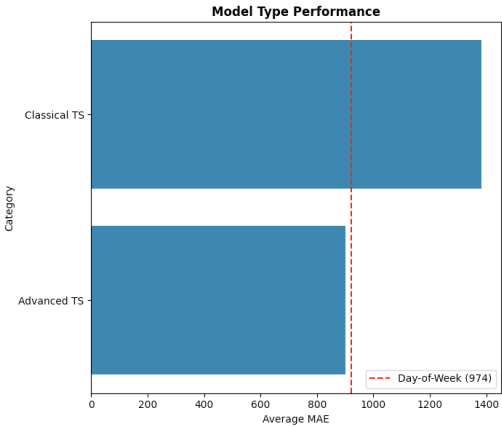
Time Series Results

===== EXECUTIVE SUMMARY =====
Best Model: Prophet
MAE: 657.7937
Improvement: +28.5%
Models Beating Benchmark: 2/6
Average MAE: 1259.4534
Performance Spread: 657.7937 - 1971.4033
Status: Success



Top 10 Model Metrics

Model	MAE	Improvement (%)	Status
Prophet	657.7937297455584	28.546038368610187	BEATS
SARIMAX	682.3825064484033	25.8750407174658	BEATS
TBATS	1286.8009285862195	-39.78093742851972	FAILS
VAR	1363.4914830942623	-48.11157922625677	FAILS
ARIMA	1594.8484745789428	-73.24312555324084	FAILS
Holt-Winters	1971.4032865075867	-114.14703184932576	FAILS



Deployment Recommendation

- HOLD: Limited improvement
- Use Day-of-Week for production
 - Improve feature engineering
 - Review data quality
 - Consider external data

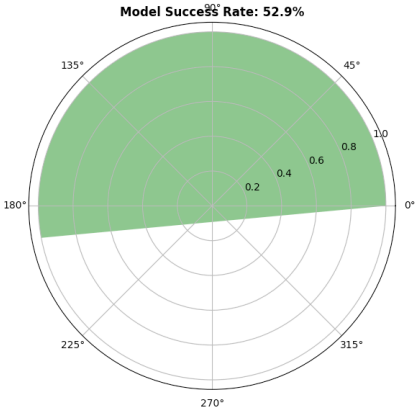
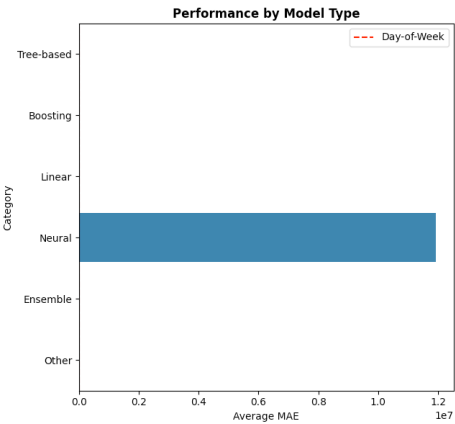
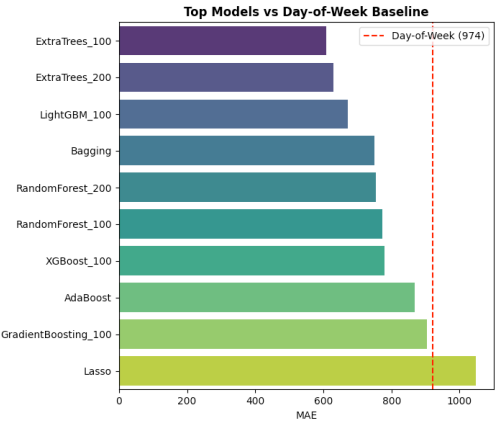
Machine Learning Results

EXECUTIVE SUMMARY

Best Model: ExtraTrees_100
MAE: 688.7227
Improvement: +33.9%
Models Beating Benchmark: 9/17
Average MAE: 1136775.8668
Performance Spread: 688.7227 - 17273510.7696
Status: ☒ Success

Top 10 Model Metrics

Model	MAE	Improvement (%)	Status
ExtraTrees_100	688.722735750737	33.87645847382525	☒ BEATS
ExtraTrees_200	628.7761707141966	11.69812002567544	☒ BEATS
LightGBM_100	671.1841571367532	7.091480439928734	☒ BEATS
Bagging	750.1286885245802	18.51599657381625	☒ BEATS
RandomForest_200	754.1523977932434	18.078914317132256	☒ BEATS
RandomForest_100	774.0187930541074	15.92089337449893	☒ BEATS
XGBoost_100	778.4741911340932	15.43691818734867	☒ BEATS
AdaBoost	867.6999825647817	5.744614978735028	☒ BEATS
GradientBoosting_100	904.7491882445808	7.200820569301394	☒ BEATS
☐ Seasonal 7Day	920.5839882453611	0.0	BASLINE



Recommendation

☒ DEPLOY: Strong ML performance